

Conditional Verification of Spectral Operator Self-Adjointness as a Pathway to the Riemann Hypothesis

Input Question

Is the Riemann hypothesis true?

Domain

Mathematical Sciences

Validation Status

needs_data

Problem Statement

The Riemann Hypothesis (RH) posits that all non-trivial zeros of the Riemann zeta function have a real part equal to one-half. This conjecture is central to understanding the distribution of prime numbers. The core scientific problem is to determine the truth value of this statement through rigorous mathematical proof or disproof, specifically by evaluating proposed theoretical frameworks such as spectral operator constructions.

Rationale

While the Riemann Hypothesis remains unproven, specific preprints propose pathways to its solution, such as the spectral operator construction in EV-Q004-d15e6434c915a4c751ae0b4e. Verifying the internal mathematical consistency and functional analytic properties (e.g., essential self-adjointness) of these proposed constructions provides a concrete, falsifiable research direction. This approach avoids asserting the truth of RH directly and instead focuses on validating the necessary conditions implied by specific theoretical models found in the allowed evidence.

Generated Hypotheses

Hypothesis 1

- Hypothesis**: The spectral operator construction in EV-Q004-d15e6434c915a4c751ae0b4e satisfies the necessary conditions (essential self-adjointness and exact spectral correspondence) to imply the Riemann Hypothesis.
- Mechanism**: According to the Hilbert-Pólya conjecture framework referenced in EV-Q004-d15e6434c915a4c751ae0b4e, if an operator D^+ is essentially self-adjoint ($D^{\dagger} = D^+$) on a defined Hilbert space domain, its spectrum must be real. If this spectrum maps bijectively to the imaginary parts of the non-trivial zeros of $\zeta(s)$ via $s = 1/2 + i\lambda$, then the reality of the eigenvalues necessitates $\text{Re}(s) = 1/2$. This hypothesis frames the verification of these specific functional analytic properties as a conditional test of the preprint's validity, rather than asserting RH truth directly.

- ****Falsifiable Prediction****: Rigorous application of von Neumann's criterion for essential self-adjointness to the operator D_+ defined in EV-Q004-d15e6434c915a4c751ae0b4e will yield non-zero deficiency indices (n_+ , $n_- \neq 0$), OR formal derivation will reveal that the spectral mapping fails to be bijective for at least one known zero, thereby invalidating the construction as a proof of RH.

- ****Required Observations****: Calculation of deficiency indices for D_+ using von Neumann's criterion or Kato-Rellich theorem ; Formal proof of domain equality $\text{Dom}(D_+) = \text{Dom}(D_+^\dagger)$; Rigorous demonstration of spectral bijection between $\sigma(D_+)$ and $\{\text{Im}(\rho) : \zeta(\rho)=0, 0 < \text{Re}(\rho) < 1\}$; Verification that boundary terms vanish in the integration by parts required for hermiticity

- ****Risk of Being Wrong****: High. The source is an unverified arXiv preprint; historical attempts at spectral proofs of RH frequently fail due to subtle domain definition errors or failure of self-adjointness. Numerical agreement for finite N does not guarantee universal validity.

Hypothesis 2

- ****Hypothesis****: The analytical strategy proposed in EV-Q004-0c6bf391479d595a3fac9466 contains a verifiable logical pathway that, if formally validated, would establish sufficient bounds to exclude off-critical-line zeros.

- ****Mechanism****: The preprint outlines a sequence of analytic number theory estimates intended to constrain zero locations. This hypothesis posits that the internal logic of this specific document constitutes a complete, gap-free deduction chain. Verification requires auditing each lemma against established unconditional results to determine if the claimed bounds are mathematically sound and sufficient to prove RH, treating the document as a candidate formal object rather than established fact.

- ****Falsifiable Prediction****: A step-by-step formal audit of EV-Q004-0c6bf391479d595a3fac9466 will identify at least one inequality or estimate that contradicts known unconditional results in analytic number theory, or reveals a logical gap/circularity that prevents the exclusion of zeros with $\text{Re}(s) \neq 1/2$.

- ****Required Observations****: Step-by-step validation of all inequalities against standard analytic number theory literature ; Identification of any unproven assumptions or circular reasoning within the strategy document ; Comparison of derived bounds with known zero-free regions to check for consistency

- ****Risk of Being Wrong****: Very High. The preprint dates from 2002 and lacks peer-reviewed acceptance, strongly suggesting undetected errors or incompleteness despite the explicit claim of a 'strategy for proving'.

Hypothesis 3

- ****Hypothesis****: Current allowed evidence provides insufficient mathematical content to formulate a valid conditional verification hypothesis for RH beyond identifying specific unverified proposals.

- ****Mechanism****: This meta-hypothesis asserts that EV-Q004-d15e6434c915a4c751ae0b4e and EV-Q004-0c6bf391479d595a3fac9466 lack the rigorous definitions necessary for formal verification, while EV-Q004-1d3b5837d65eed2070027b37 and EV-Q004-bbce4f29714512c37c5e4088 are topically irrelevant. Thus, no testable research plan can be generated without external knowledge.

- ****Falsifiable Prediction****: If detailed content analysis of EV-Q004-d15e6434c915a4c751ae0b4e reveals a sufficiently precise definition of operator D_+ and its domain to permit calculation of deficiency indices, this hypothesis is falsified.

- **Required Observations**: Content verification confirming whether EV-Q004-d15e6434c915a4c751ae0b4e defines D^+ precisely enough for functional analytic testing ; Confirmation of topical irrelevance for EV-Q004-1d3b5837d65eed2070027b37 and EV-Q004-bbce4f29714512c37c5e4088
- **Risk of Being Wrong**: Medium. Risk exists if the preprints contain hidden precision not captured in current extraction metadata.

Technical Details

This research plan addresses the conditional verification of the spectral operator construction proposed in EV-Q004-d15e6434c915a4c751ae0b4e. The hypothesis does not assert the truth of the Riemann Hypothesis (RH) but rather tests whether the specific operator D^+ defined in the preprint satisfies the rigorous functional analytic conditions (essential self-adjointness and exact spectral bijection) required to imply RH under the Hilbert-Pólya framework. The verification strategy involves: (1) Formal definition of the Hilbert space H and operator D^+ domain; (2) Application of von Neumann's criterion to calculate deficiency indices (n_+, n_-) to determine essential self-adjointness; (3) Rigorous proof of spectral bijection between $\sigma(D^+)$ and the imaginary parts of non-trivial zeta zeros. Numerical checks of eigenvalues are strictly limited to heuristic debugging for falsification purposes and do not constitute proof. The primary validation method is formal verification using proof assistants (Lean/Coq) or explicit analytic derivation, replacing any LLM-based peer review simulation.

Datasets

Source

```
```json
[
{
 "id": "EV-Q004-d15e6434c915a4c751ae0b4e",
 "description": "ArXiv preprint math/0102031 defining the spectral operator D^+ and its construction details.",
 "type": "theoretical_framework"
}
]
```

### Target

Formal mathematical verification of essential self-adjointness and spectral correspondence properties of  $D^+$ .

## Paper Abstract

Background: The Riemann Hypothesis remains one of the most significant unsolved problems in mathematics, relating the distribution of prime numbers to the zeros of the Riemann zeta function. While unproven, various

theoretical frameworks, such as the Hilbert-Pólya conjecture, suggest connections to spectral theory. Methods: This study proposes a rigorous verification plan for the spectral operator construction outlined in EV-Q004-d15e6434c915a4c751ae0b4e. We focus on testing the essential self-adjointness of the operator  $D_+$  and the bijectivity of its spectral mapping to zeta zeros. Validation Plan: We employ von Neumann's criterion for deficiency indices and formal verification tools (Lean/Coq) to audit the logical consistency of the operator's domain and boundary conditions. Results: pending. The current status indicates that experimental verification and formal proof auditing are yet to be executed.

## Methods

1. Functional Analytic Verification: Extract the precise definition of operator  $D_+$  and its domain from EV-Q004-d15e6434c915a4c751ae0b4e. Apply von Neumann's criterion to compute deficiency indices. 2. Spectral Bijection Proof: Formally derive the mapping between the spectrum of  $D_+$  and the zeros of  $\zeta(s)$ . 3. Boundary Term Analysis: Explicitly calculate boundary terms in the integration by parts for  $\langle \phi, D_+ \psi \rangle$  to ensure hermiticity. 4. Formal Verification: Encode the core logical steps into a proof assistant (e.g., Lean or Coq) to check for logical consistency.

## Experiments

### Baselines

```
```json
[
  "Standard Self-Adjoint Extensions: Compare  $D_+$  against known self-adjoint extensions of similar differential operators to identify if  $D_+$  requires fine-tuned boundary conditions that break generality.",
  "Known Zero-Free Regions: Verify that the spectral properties of  $D_+$  do not contradict established unconditional results regarding zero-free regions of  $\zeta(s)$ .",
  "Hardy's Theorem Consistency: Ensure the operator construction respects the existence of infinitely many zeros on the critical line as proven by Hardy."
]
```
```

### Metrics

```
```json
[
  "Deficiency Indices Status: Binary outcome ( $n_+=n_-=0$  vs. non-zero) determining essential self-adjointness.",
  "Spectral Bijection Validity: Binary pass/fail on the rigorous proof of one-to-one correspondence between  $\sigma(D_+)$  and  $\text{Im}(\rho)$ .",
  "Boundary Anomaly Count: Integer count of non-vanishing boundary terms; must be zero for hermiticity.",

```

"Formal Proof Completion Rate: Percentage of logical steps successfully verified in Lean/Coq without holes."

]

...

Ablation

Test the robustness of self-adjointness by varying the domain definition slightly to see if the property holds only under exact, fragile conditions specified in the preprint.

Validation Protocol

Step-by-step formal verification using automated theorem proving assistants (Lean/Coq). Manual expert audit of the functional analytic derivations, specifically focusing on domain definitions and boundary conditions. Numerical diagonalization of truncated matrices is used ONLY for initial sanity checks and falsification heuristics, not as evidence for the universal claim.

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- **EV-Q004-d15e6434c915a4c751ae0b4e** · arxiv · arXiv:math/0102031
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/math/0102031.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:2|section:page-2|paragraph:1;
content_sha256=0b94caa5fd0050f6de8f94567dad234f368c80cdfaff4095ab175862a65a4e51

Reviewer Comments

- The revised hypothesis correctly frames the investigation as a conditional verification of the spectral operator construction in EV-Q004-d15e6434c915a4c751ae0b4e rather than asserting the truth of the Riemann Hypothesis, resolving the previous evidence grounding violation.
- Invalid 'Peer-Review Simulation' and 'Peer Consensus Index' metrics have been successfully removed and replaced with rigorous functional analytic criteria (deficiency indices, domain equality) and formal verification targets.
- Numerical spectral matching is now explicitly relegated to heuristic debugging/falsification in technical_details, preventing conflation with formal proof.
- Experiment design includes appropriate baselines (Hardy's Theorem, Zero-Free Regions) and binary formal metrics suitable for verifying mathematical properties.

- Results field correctly states 'pending' without fabrication.

Revision History

- auto_revision_1: 依据评审意见重做假设与实验设计。

Reproducibility Checklist

- Full LaTeX source code of the operator D^+ definition and domain specification from EV-Q004-d15e6434c915a4c751ae0b4e.
- Explicit statement of the Hilbert space inner product and norm.
- Detailed derivation of the adjoint operator $D^\dagger$ and calculation of deficiency indices.
- List of all assumptions regarding boundary behavior at infinity and singular points.
- Lean/Coq formalization scripts for the core logical steps of the self-adjointness proof.